

## RS002 Week 12 — English Worksheet

**Topic:** Refactor the Whole App into Functions · **Course:** Pokédex App · **Time:** about 45 minutes

This week you think about how a whole app gets tidied up. All those lines written one by one over the last 11 weeks get grouped into **named functions**, and the main code shrinks to a short **menu loop** that just *calls* them. This is called **refactoring** — same behaviour, cleaner shape. On this worksheet you read code, predict it, fix it on paper, and explain the code you wrote in the live lesson.

Words to know: **refactor**, **function**, **menu loop**, **call**, **break**

### 1 · Recall

From memory:

**Write a function** `count_dex(pokedex)` **that returns how many Pokémon are in the list.**

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## 2 · Predict 🧠

Read each block. Write what you think will happen — just from reading, no running.

```
def show_menu():
    print("1. 도감 보기")
    print("2. 종료")

show_menu()
choice = input("선택: ").strip()
print(f"고른 메뉴: {choice}")
```

The user types **1**. What is printed in total?

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```
while True:
    cmd = input("명령 (q로 종료): ").strip()
    if cmd == "q":
        break
    print(f"입력: {cmd}")
```

The user types **a**, then **b**, then **q**. What does each round do, and when does it stop?

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```
trainer = input("이름: ").strip() or "이름 모를 트레이너"
print(trainer)
```

The user presses Enter without typing. What prints? What does `or` do here?

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### 3 · Spot the Bug 🐛

Read what each block is **supposed** to do, fix it on paper, then explain the fix.

**Bug A** — A function must be **defined above** where it is called. Right now `show_menu` is called before it exists, so the program crashes.

```
show_menu()

def show_menu():
    print("=== 메뉴 ===")
```

**Hint:** move the definition above the call.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug B** — This menu loop should keep running until the user picks `3`. Right now it runs once and stops.

```
choice = input("선택: ").strip()
if choice == "1":
    print("도감 보기")
elif choice == "3":
    print("종료")
```

**Hint:** wrap it in `while True:` and `break` on `3`.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug C** — When the user types something that is not a menu number, the app should say so. Right now an unknown choice silently does nothing.

```
while True:
    choice = input("선택: ").strip()
    if choice == "1":
        print("도감 보기")
    elif choice == "2":
        break
```

**Hint:** add an `else` for unknown input.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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## 4 · Explain the Code

Read this working function-based app, then answer the questions below.

```

def show_intro(trainer_name):
    print(f"\n=== 포켓몬 도감 ===")
    print(f"환영합니다, {trainer_name} 트레이너님!\n")

def show_menu():
    print("\n--- 메뉴 ---")
    print("1. 도감 보기")
    print("2. 포켓몬 검색")
    print("3. 종료")

def list_all(pokedex):
    for p in pokedex:
        print(f"  {p['name']} 타입:{p['type']} HP:{p['hp']}")

def search_pokemon(pokedex):
    keyword = input("검색어: ").strip().lower()
    matches = [p for p in pokedex if keyword in p["name"].lower()]
    if matches:
        for p in matches:
            print(f"  {p['name']} ({p['type']})")
    else:
        print("검색 결과 없음")

pokedex = [
    {"name": "피카츄", "type": "전기", "hp": 35},
    {"name": "이상해씨", "type": "풀", "hp": 45},
]

trainer = input("트레이너 이름: ").strip() or "이름 모를 트레이너"
show_intro(trainer)

while True:
    show_menu()
    choice = input("선택: ").strip()
    if choice == "1":
        list_all(pokedex)
    elif choice == "2":
        search_pokemon(pokedex)
    elif choice == "3":
        print("또 만나요!")
        break
    else:
        print("1~3 중에서 골라주세요.")

```

What does the `while True:` loop do, and which line stops it?

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When the user types `1`, which function gets called, and what does it print?

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Why is each action (intro, menu, list, search) written as its own function instead of all in the loop?

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`search_pokemon` builds `matches` with a list comprehension. In your own words, what ends up inside `matches` ?

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What happens if the user types `7`, and which line handles that?

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## 5 · Explain Your Lesson Code 🎥

In the live lesson you refactored your Pokédex app into functions. Make a short phone video explaining the code **you** wrote today. You may show it running. Try to use: **refactor**, **function**, **menu loop**, **call**, **break**.

1. Run your app and try every menu option, including quit.
2. Scroll to your main loop and show how short it is.
3. Point to one of your functions and explain what the loop *calls*.
4. Say why grouping code into functions made it easier to read and fix.

**Write what you will say in your video. Plan it here before you record.**

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**Submit** 

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.